

Problem

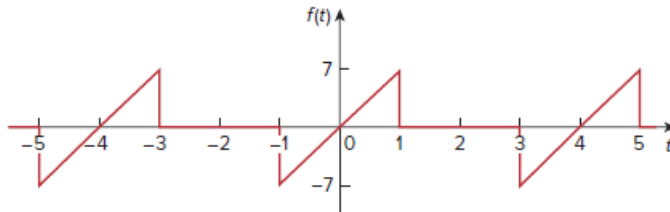
For the waveform shown in Fig. 17.65 below,

(a) specify the type of symmetry it has.

(b) calculate a_3 and b_3 ,

(c) find the rms value using the first five nonzero harmonics.

Figure 17.65:



Step-by-step solution

Step 1 of 10

Refer to Equation 17.3 in the textbook.

The Fourier series is,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Where,

$$\omega_0 = \frac{2\pi}{T} \text{ is the fundamental frequency.}$$

The function is,

$$f(t) = \begin{cases} 7t, & 0 < t < 1 \\ 0, & 1 < t < 3 \\ 7(t-4), & 3 < t < 4 \end{cases}$$

Step 2 of 10

(a)

Since $f(-t) = -f(t)$, the function is having odd symmetry.

Step 3 of 10

The time period is, $T = 4$.

Determine the value of ω_0 .

$$\begin{aligned} \omega_0 &= \frac{2\pi}{4} \\ &= \frac{\pi}{2} \end{aligned}$$

Refer to Equation 17.28 in the textbook.

Since the function is odd symmetric,

$$a_0 = 0$$

$$a_n = 0$$

Determine the value of b_n .

$$\begin{aligned} b_n &= \frac{4}{T} \int_0^{T/2} f(t) \sin(n\omega_0 t) dt \\ &= \frac{4}{4} \int_0^2 f(t) \sin(n\omega_0 t) dt \\ &= \int_0^1 f(t) \sin(n\omega_0 t) dt + \int_1^2 f(t) \sin(n\omega_0 t) dt \\ &= \int_0^1 7t \sin\left(\frac{n\pi}{2}t\right) dt + \int_1^2 0 \cdot \left(\frac{n\pi}{2}t\right) dt \\ &= 7\left(\frac{2}{n\pi}\right) \left[-t \cos\left(\frac{n\pi}{2}t\right) \right]_{t=0}^{t=1} + 7\left(\frac{2}{n\pi}\right)^2 \left[\sin\left(\frac{n\pi}{2}t\right) \right]_{t=0}^{t=1} \\ &= 0 + \frac{28}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right) - \frac{14}{n\pi} \cos\left(\frac{n\pi}{2}\right) \\ &= \frac{28}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right) - \frac{14}{n\pi} \cos\left(\frac{n\pi}{2}\right) \end{aligned}$$

Step 5 of 10

(b)

Determine the value of a_3 .

$$a_3 = 0 \text{ (Since odd symmetry)}$$

Therefore, the value of a_3 is .

Step 6 of 10

Determine the value of b_3 .

$$\begin{aligned} b_3 &= \frac{28}{(3)^2 \pi^2} \sin\left(\frac{3\pi}{2}\right) - \frac{14}{3\pi} \cos\left(\frac{3\pi}{2}\right) \\ &= -\frac{28}{9\pi^2} \\ &= -0.3152 \end{aligned}$$

Therefore, the value of b_3 is .

Step 7 of 10

(c)

Determine the rms value.

$$F_{\text{rms}} = \sqrt{a_0^2 + \sum_{n=1}^{\infty} \frac{1}{2} (a_n^2 + b_n^2)}$$

Determine the value of b_1 .

$$\begin{aligned} b_1 &= \frac{28}{(1)^2 \pi^2} \sin\left(\frac{\pi}{2}\right) - \frac{14}{\pi} \cos\left(\frac{\pi}{2}\right) \\ &= 2.837 \end{aligned}$$

Step 8 of 10

Determine the value of b_2 .

$$\begin{aligned} b_2 &= \frac{28}{4\pi^2} \sin\left(\frac{2\pi}{2}\right) - \frac{14}{2\pi} \cos\left(\frac{2\pi}{2}\right) \\ &= 2.2282 \end{aligned}$$

Determine the value of b_3 .

$$\begin{aligned} b_3 &= \frac{28}{(3)^2 \pi^2} \sin\left(\frac{3\pi}{2}\right) - \frac{14}{3\pi} \cos\left(\frac{3\pi}{2}\right) \\ &= -\frac{28}{9\pi^2} \\ &= -0.3152 \end{aligned}$$

Step 9 of 10

Determine the value of b_4 .

$$\begin{aligned} b_4 &= \frac{28}{16\pi^2} \sin\left(\frac{4\pi}{2}\right) - \frac{14}{4\pi} \cos\left(\frac{4\pi}{2}\right) \\ &= -1.1141 \end{aligned}$$

Determine the value of b_5 .

$$\begin{aligned} b_5 &= \frac{28}{25\pi^2} \sin\left(\frac{5\pi}{2}\right) - \frac{14}{4\pi} \cos\left(\frac{5\pi}{2}\right) \\ &= 0.1135 \end{aligned}$$

Step 10 of 10

Determine the rms value of first five nonzero harmonics.

$$\begin{aligned} F_{\text{rms}} &= \sqrt{0 + \sum_{n=1}^5 \frac{1}{2} (b_n^2)} \\ &= \sqrt{0 + \sum_{n=1}^5 \frac{1}{2} (b_n^2)} \\ &= \sqrt{0 + \sum_{n=1}^5 \frac{1}{2} (2.837^2 + 2.2282^2 + (-0.3152)^2 + (-1.1141)^2 + 0.1135^2)} \\ &= 2.6802 \end{aligned}$$

Therefore, the rms value of the periodic function is 2.6802.